A Modern Handbook of Physics

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Preface

The basic sciences and physics, in particular, are of prime importance today in the training of engineers for the various branches of the national economy. This has led to fundamental changes in recent years in the teaching of physics in engineer ing institutes, and to the students of other educational institutions in which they do not major in physics. The scope and scientific level of physics courses have been substantially supplemented and cover the main trends in the development of modern physics. Consequently, the physics textbooks for engineering students have inevitably become three-volume editions of almost fifteen hundred pages. The need has arisen, in this connection, for a concise handbook on this subject. The aim of the authors was to fulfill this need.

In scope and depth this handbook includes all of the definitions, formulas and information covered in the most comprehensive and up-to-date physics courses of engineering institutes and the physics departments of universities and colleges. Physical laws are concisely formulated, all the necessary explanations are given and, in many cases, derivations as well. Though it plays a vital role in the teaching

of physics, experimental material has been omitted. This is due only to a lack of space. All the units and symbols comply with the requirements of the SI. Units of physical quantities and systems of units are listed and dealt with in a short appendix.

This handbook is designed primarily for engineering students, as well as college and university students studying, but not majoring in physics. It can be used to advantage by engineers and graduate students, as well as by instructors and lecturers of intermediate schools and colleges.

Mathematical knowledge required in using the handbook is within the scope of the ordinary mathematics courses of engineering institutes. The detailed index and the numerous cross references, indicating the chapter, section and subsection, are of aid in finding any required information. The authors are sincerely grateful to the translator, N. Weinstein, and the science editor, T. Baranovskaya, for their comprehensive and expert work in the publication of the English edition of this handbook, as well as for their valuable suggestions, which should facilitate its use by the reader.

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Parts One, Four, Five and the appendices were written by Prof. A. A. Detlaf; Parts Two, Three, Six, Seven and Eight, by Prof. B. M. Yavorsky.

Part I

Mechanics

Chapter 1

Kinematics

1.1 Mechanical Motion. The Subject Matter of Mechanics

1.1.1

The simplest form of motion in nature is *mechanical motion*. It consists of changes in the relative positions of bodies or their parts in space in the course of time. The branch of physics dealing with the laws of mechanical motion is called *mechanics*. In a more narrow sense of the word, this is often under stood to mean *classical mechanics*, which deals with the motion of macroscopic bodies travelling at velocities a great many times less than that of light in free space. Classical mechanics is based on Newton's laws and is therefore frequently referred to as *Newtonian mechanics*. The laws of motion of bodies at velocities close to that of light in free space are taken up in *relativistic mechanics* (**??**), and the laws

of motion of micro particles (for example, the electrons in atoms, molecules, crystals, etc.) are the subject matter of *quantum mechanics* (**??**).

1.1.2

Classical mechanics comprises three basic branches: statics, kinematics and dynamics. Statics deals with the laws for the composition of forces and the conditions for the equilibrium of bodies. Kinematics provides a mathematical description of all possible kinds of mechanical motion, regardless of what causes each particular kind of motion. Dynamics investigates the effect of the interaction between bodies on their mechanical motion.

1.1.3

The mechanical properties of bodies are determined by their chemical nature, internal structure and state. These are beyond the scope of mechanics and are studied in other branches of physics. Hence, in accordance with the conditions of each specific problem on hand, various simplified models are employed to describe real bodies in motion. These include the particle, the perfectly rigid body, the perfectly elastic body, the perfectly inelastic body and others.

A material point, or particle, is a body whose shape and size are of no consequence under the conditions of the problem being considered. For example, in a first approximation, the motion of a ship from one point to another can be dealt with as the motion of a particle. If, however, it becomes necessary to take into account such details of this motion as the pitching and rolling of the ship by swells or heavy seas, the ship should be dealt with as an extended body of

definite shape. In the literature the word particle is more frequently used than material point.

Any extended body or system of such bodies, making up the mechanical system being investigated, can be regarded as a system of particles. For this purpose, ail the bodies of the system can be conceived of as being broken down into such a large number of parts that the size of each part is negligibly small compared to the sizes of the bodies themselves.

1.1.4

A perfectly rigid body, or simply rigid body, is one whose deformation can be neglected under the conditions of the given problem. The distance between any two points of a rigid body remains constant no matter what action the body is subjected to. A perfectly rigid body can be regarded as a system of particles rigidly connected to one another.

A perfectly elastic body is one whose deformation obeys Hooke's law (**??**). After the external forces applied to such a body have been removed, its initial size and shape are completely restored.

A perfectly inelastic body is one that completely retains its deformed state, caused by external forces applied to the body, when these forces have been removed.